HEAP Priority Queue
(FAANG Cheat Sheet)

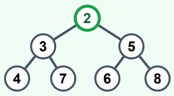
1. WHAT IS A HEAP?

- A Heap is a **Complete Binary Tree**.
- It satisfies the **Heap Property**.
- It is **NOT** a sorted data structure.
- Only the root element is guaranteed to be the minimum (Min Heap) or maximum (Max Heap).
- Heaps are used to implement **Priority Queue**.

2. MIN HEAP VS MAX HEAP

MIN HEAP

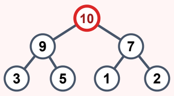
Parent ≤ Children



Smallest element is at the root.

MAX HEAP

Parent ≥ Children



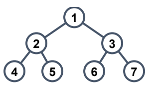
Largest element is at the root.

3. COMPLETE BINARY TREE

All levels are completely filled except possibly the last level, and all nodes in the last level are as far left as possible.

4. ARRAY REPRESENTATION (0-INDEXED)

Example Tree



Index Relationships

Parent(i) = (i - 1) / 2
Left(i) = 2 * i + 1
Right(i) = 2 * i + 2

Array Representation

Index :	0	1	2	3	4	5	6
Value :	1	2	3	4	5	6	7

5. TIME COMPLEXITIES

Operation	Min Heap	Max Heap	Why?
offer(x)	O(log n)	O(log n)	Heapify Up
poll()	O(log n)	O(log n)	Heapify Down
peek()	O(1)	O(1)	Root access
size()	O(1)	O(1)	Just length
isEmpty()	O(1)	O(1)	Just length
Build Heap (n elements)	O(n)	O(n)	Bottom-up heapify

⭐ **Build Heap** (Bottom-Up) is O(n) which is better than inserting n elements one by one (O(n log n)).

6. JAVA PRIORITYQUEUE

a) Min Heap (Default)

```java
PriorityQueue<Integer> minHeap = new PriorityQueue<>();
// Smallest element at peek()
```

b) Max Heap (3 Ways)

1. Using Collections.reverseOrder() (Recommended)

```java
PriorityQueue<Integer> maxHeap =
    new PriorityQueue<>(Collections.reverseOrder());
```

2. Using Comparator (Integer.compare)

```java
PriorityQueue<Integer> maxHeap =
    new PriorityQueue<>((a, b) -> Integer.compare(b, a));
```

3. Using subtraction (**Not recommended** – overflow risk)

```java
PriorityQueue<Integer> maxHeap =
    new PriorityQueue<>((a, b) -> b - a);
```

c) Common Operations

Method	Description
`offer(x)`	Inserts element x into heap.
`poll()`	Removes and returns root element (min or max).
`peek()`	Returns root element without removing.
`size()`	Returns number of elements.
`isEmpty()`	Checks if heap is empty.

7. IMPORTANT TERMINOLOGY

Root : Topmost element.
Parent(i) : (i - 1) / 2
Left Child(i) : 2 * i + 1
Right Child(i) : 2 * i + 2
Height : Longest path from root to leaf.
Leaf Node : Node with no children.
Heapify : Process of restoring heap property (Bubble Up or Bubble Down).

8. KEY POINTS TO REMEMBER

- ✓ Heap is a Complete Binary Tree.
- ✓ Heap ≠ Sorted Array.
- ✓ Heaps are used for **Priority**, not order.
- ✓ Choose **Min Heap** or Max Heap based on the problem goal.
- ✓ Heap operations are logarithmic time (O(log n)).

9. COMMON MISTAKES

- ✗ Confusing Min Heap with Max Heap.
- ✗ Using wrong comparator.
- ✗ Forgetting that default PriorityQueue in Java is Min Heap.
- ✗ Not handling empty heap before poll/peek.
- ✗ Assuming heap gives sorted order.

💡 **Golden Rule:** Heaps are all about **PRIORITY**. Ask: *"What element should come out first?"*

→ Next Page: Heap Patterns Overview
(When to use which heap & pattern summary)

HEAP – PATTERNS

Top Interview Patterns with Heaps (Priority Queue)

Max Heap → largest element on top **Min Heap → smallest element on top** **PQ → PriorityQueue**

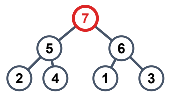
PATTERN 1 REPEATED MAX / MIN RETRIEVAL (e.g., Last Stone Weight)

WHEN TO USE?

- We always take the largest (or smallest) element.
- Do something with it.
- Put the result back.
- Repeat until 1 (or none) left.

HEAP TO USE

Max Heap → largest



IDEA / AHA

We always care about the current largest element.

Heap gives it in O(1).

TEMPLATE (JAVA)

```java
PriorityQueue<Integer> pq =
  new PriorityQueue<>(Collections.
  reverseOrder()); // Max Heap

// Build heap
for (int x : arr) pq.offer(x);

while (pq.size() > 1) {
  int a = pq.poll(); // largest
  int b = pq.poll(); // 2nd largest
  if (a != b) pq.offer(a - b);
}
return pq.isEmpty() ? 0 : pq.peek();
```

COMPLEXITY

Time: O(n log n) (n insertions + up to n polls)
Space: O(n)

Examples
- LeetCode 1046 Last Stone Weight

PATTERN 2 TOP-K ELEMENTS (e.g., Kth Largest, Top K Frequent)

WHEN TO USE?

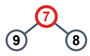
- Need the K largest/smallest elements.
- K most frequent, top K, K closest, etc.
- We discard the rest.

HEAP TO USE

Top K Largest → Min Heap (size K)

(Keep K=3 largest seen so far)

Top K Smallest → Max Heap (size K)



(Keep K=3 smallest seen so far)

IDEA / AHA

Keep only K best candidates in heap.

If better element comes, replace root (worst among K).

TEMPLATE (JAVA) → TOP K LARGEST (MIN HEAP)

```java
PriorityQueue<Integer> pq = new PriorityQueue<>();

for (int x : nums) {
  pq.offer(x);
  if (pq.size() > k) pq.poll(); // remove smallest
}
// pq contains top K largest
List<Integer> res = new ArrayList<>(pq);
```

Top K Smallest (Max Heap)

```java
PriorityQueue<Integer> pq = new PriorityQueue<>(
  Collections.reverseOrder());

for (int x : nums) {
  pq.offer(x);
  if (pq.size() > k) pq.poll(); // remove largest
}
// pq contains top K smallest
```

COMPLEXITY

Time: O(n log k)
Space: O(k)

Examples
- LC 215 Kth Largest Element in Array
- LC 347 Top K Frequent Elements
- LC 973 K Closest Points to Origin
- LC 692 Top K Frequent Words

PATTERN 3 MERGE K SORTED LISTS / ARRAYS (e.g., Merge K Sorted Lists)

WHEN TO USE?

- We have K sorted lists/arrays.
- Need to merge into one sorted output.
- Always pick the smallest current candidate.

HEAP TO USE

Min Heap

1→4→7
2→5→8
3→6→9
↓
Min Heap
↓
1 2 3 4 5 6 7 8 9

IDEA / AHA

Put the first element of each list into heap.

Pop smallest node, add its next node (if exists).

TEMPLATE (JAVA)

```java
class Node { int val; Node next; }

PriorityQueue<Node> pq = new PriorityQueue<>(
  (a, b) -> a.val - b.val); // Min Heap

// 1. Add head of each list
for (Node head : lists)
  if (head != null) pq.offer(head);

Node dummy = new Node(0), tail = dummy;

while (!pq.isEmpty()) {
  Node cur = pq.poll();
  tail.next = cur; tail = tail.next;
  if (cur.next != null) pq.offer(cur.next);
}
return dummy.next;
```

COMPLEXITY

Time: O(N log k) (N = total nodes, k = number of lists)
Space: O(k)

Examples
- LeetCode 23 Merge K Sorted Lists
- LeetCode 373 Find K Pairs with Smallest Sums

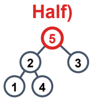
PATTERN 4 TWO HEAPS (MEDIAN FROM DATA STREAM) (e.g., Find Median from Data Stream)

WHEN TO USE?

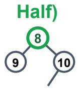
- Need median at any point in a stream of numbers.
- Insertions happen one by one.
- Need O(log n) insert and O(1) median.

HEAP TO USE

Max Heap (Smaller Half)



Min Heap (Larger Half)



IDEA / AHA

Maintain two halves.

1) maxHeap.peek() ≤ minHeap.peek()
2) Size difference ≤ 1

Median is either top of

TEMPLATE (JAVA) - (ADD NUMBER)

```java
// maxHeap: left half (max heap)
// minHeap: right half (min heap)
PriorityQueue<Integer> maxHeap =
  new PriorityQueue<>(Collections.reverseOrder());
PriorityQueue<Integer> minHeap = new PriorityQueue<>();

void addNum(int num) {
  if (maxHeap.isEmpty() || num <= maxHeap.peek())
    maxHeap.offer(num);
  else
```

COMPLEXITY

addNum: O(log n)
findMedian: O(1)
Space: O(n)

Examples

		one heap or avg of both.	<pre> minHeap.offer(num);
			  // Rebalance
			  if (maxHeap.size() > minHeap.size() + 1)
			    minHeap.offer(maxHeap.poll());
			  else if (minHeap.size() > maxHeap.size())
			    maxHeap.offer(minHeap.poll());
			}
			double findMedian() {
			  if (maxHeap.size() == minHeap.size())
			    return (maxHeap.peek() + minHeap.peek()) / 2.0;
			  return maxHeap.peek();
			}</pre>
				• LeetCode 295 Find Median from Data Stream

⭐ AHA SUMMARY

✓ What do I care about most right now? Put it on top!
✓ Use heap to discard what I don't need.
✓ Heap ≠ Sorting. Heap gives priority, not full order.

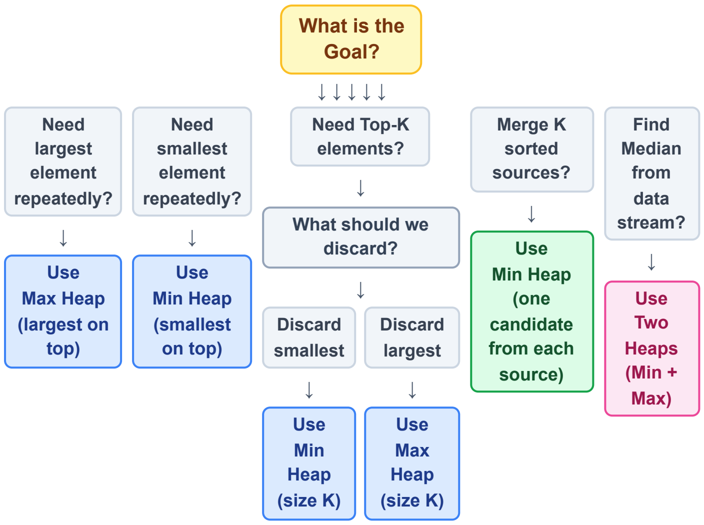
🔴 PATTERN RECOGNITION TRIGGERS

Top K / Kth / Most Frequent / Closest / Smallest / Largest / Merge K / Running Median / Priority / Stream / Scheduler

HEAP – CHEAT SHEET & DECISION GUIDE

Quick Reference for Interviews

1. HEAP DECISION TREE



2. JAVA TEMPLATES (QUICK)

Min Heap (Default)

```java
PriorityQueue<Integer> minHeap =
    new PriorityQueue<>();
// smallest element at peek()
```

Max Heap (Using reverseOrder)

```java
PriorityQueue<Integer> maxHeap =
    new PriorityQueue<>(Collections.reverseOrder());
// largest element at peek()
```

Max Heap (Using Comparator)

```java
PriorityQueue<Integer> maxHeap =
    new PriorityQueue<>((a, b) -> Integer.compare(b, a));
// largest element at peek()
```

Top-K Largest (Min Heap of size K)

```java
PriorityQueue<Integer> pq = new PriorityQueue<>();
for (int x : nums) {
    pq.offer(x);
    if (pq.size() > k) pq.poll(); // remove smallest
}
// pq contains k largest
```

Top-K Smallest (Max Heap of size K)

```java
PriorityQueue<Integer> pq =
    new PriorityQueue<>((a, b) -> Integer.compare(b, a));
for (int x : nums) {
    pq.offer(x);
    if (pq.size() > k) pq.poll(); // remove largest
}
// pq contains k smallest
```

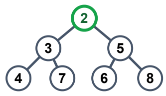
3. COMPLEXITY TABLE

Operation	Min Heap	Max Heap	Notes
Insert (offer)	O(log n)	O(log n)	Heapify Up
Remove (poll)	O(log n)	O(log n)	Heapify Down
Peek	O(1)	O(1)	Root access
Size	O(1)	O(1)	Just length
isEmpty	O(1)	O(1)	Just length
Build Heap (from n elements)	O(n)	O(n)	Bottom-up heapify

4. KEY INVARIANTS

Min Heap

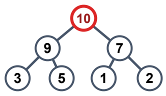
```
For every node i:
parent(i) <= child(i)
Root = smallest element
```



Always true for all nodes

Max Heap

```
For every node i:
parent(i) >= child(i)
Root = largest element
```



Always true for all nodes

5. COMMON INTERVIEW CHECKLIST

- ✓ Understand the goal clearly.
- ✓ Ask: What should I keep? What should I discard?
- ✓ Choose the right Heap (Min / Max).
- ✓ Decide the size (K or all elements).
- ✓ Maintain Heap invariant after every insertion/removal.
- ✓ Think about edge cases (empty, k = 0, k > n, duplicates).
- ✓ Analyze Time and Space Complexity.

6. COMMON MISTAKES

- ✗ Using Min Heap when Max Heap is required (or vice versa).
- ✗ Forgetting that PriorityQueue in Java is Min Heap by default.
- ✗ Not maintaining heap size while solving Top-K problems.
- ✗ Forgetting to add next element in Merge K after poll().
- ✗ Not handling empty heap before peek()/poll().
- ✗ Using subtraction in comparator (a - b) → overflow risk.
- ✗ Confusing Heap with Sorted Array / Tree.

7. AHA MOMENTS

- ⭐ Heap gives you the best element in O(1).
- ⭐ It is about PRIORITY, not sorting.
- ⭐ Think: "What should I discard?" for Top-K problems.
- ⭐ Merge K = One candidate per source.
- ⭐ Median = Balance two halves.

🎯 Pattern Recognition Triggers

- → Top K / Kth Largest / Kth Smallest
- → Most Frequent / Least Frequent
- → Merge K Sorted / K Sorted Lists
- → Find Median / Running Median
- → Task Scheduler / CPU Scheduling
- → Closest K / Smallest Range / Meeting Rooms

⭐ **Remember: Every Heap problem belongs to one of the 4 patterns on Page 2.**

HEAP – FAANG HIDDEN TRIGGERS

How Interviewers Hide Heap Problems (Recognition Guide)

They won't say "Use Heap"
You must recognize it !

1. WHEN TO THINK HEAP? (UNSEEN QUESTION CLUES)

	Question Says...	Think Heap Because...
🅺	Top K / K largest / K smallest	We only care about best K elements.
🎯	K closest / K farthest	Keep best K based on distance / value.
🅺th	Kth largest / Kth smallest	Heap gives Kth in optimal time.
📊	Most frequent / Top frequent	Use heap on frequency map.
∽	Continuously receive data" (data stream")	Need to maintain running result.
⚖	Running median / Median at any time	Two Heaps maintain middle efficiently.
🔗	Merge K sorted lists / arrays / streams	Pick smallest current from K sources.
🔄	Always need smallest / largest repeatedly	Heap gives min/max in O(log n) each time.
📅	Scheduling / Highest priority task	Always process highest (or lowest) priority first.
📦	Resource allocation	Allocate best resource (min waste / min cost).
🔍	Find N smallest among millions	Min Heap of size N.
🏆	Live leaderboard / Top scores	Maintain top K scores dynamically.
🪟	Sliding window maximum/minimum	(Usually Deque) but if K is large and needs frequent removal by value → Heap (advanced).

2. FAANG FAVORITE HEAP PROBLEM THEMES

System Design Like Problems — Heap (Max)
Design Twitter Feed, Task Scheduler, Rate Limiter, Job Scheduler

Streaming / Real Time Data — Two Heaps
Running Median, Data Stream as Disjoint Intervals

Top K / Ranking — Min Heap (size K)
Top K Frequent, Top K Elements, K Closest Points

Merge / K Sorted Sources — Min Heap
Merge K Sorted Lists, Smallest Range in K Lists

Greedy + Priority — Max Heap
IPO, Find K Pairs with Smallest Sums, Maximize Capital

Intervals / Meetings — Min Heap
Meeting Rooms (III), Employee Free Time, Conference Room Scheduler

Graphs / Advanced — Min Heap
Dijkstra's Algorithm, Prim's MST, Network Delay Time

💡 **Pro Tip**
Translate the problem into:
"Which element should come out first?"
→ That is your heap priority.

3. HEAP VS OTHER DS (CHOOSE RIGHT TOOL)

Problem Type / Goal	Best Tool	Why?
Get min / max repeatedly	**Heap**	O(log n) per operation
Top K elements	**Heap**	O(n log k) better than sorting
Kth largest / smallest	**Heap / Quickselect**	Heap: O(n log k), Quickselect: O(n) avg
Merge K sorted things	**Heap**	One candidate per source
FIFO order	**Queue**	Heap not for order
LIFO order	**Stack**	Heap not for order
Need sorted output	**Sort**	Full order required
Range queries	**Segment Tree / BIT**	Heap not for range aggregation
Dynamic min with delete by value	**TreeMap / Balanced BST**	Heap can't delete arbitrary element efficiently

4. INTERVIEWER TRICKS TO WATCH OUT

- ✗ They never say "Use Heap" explicitly.
- ✗ They mix multiple concepts (e.g., Heap + HashMap).
- ✗ They change the goal (e.g., maximize instead of minimize).
- ✗ They give huge constraints to hint suboptimal solutions will TLE.
- ✗ They ask for real time / online processing → think Heap.
- ✗ They change K during the process (dynamic K).
- ✗ They ask for both min and max → think Two Heaps.

5. RECOGNITION SHORTCUT (3 QUESTIONS)

1. What am I optimizing? (min/max/priority)
2. Do I need only top/bottom K or keep discarding irrelevant elements?
3. Is the data coming continuously or from multiple sorted sources?

If YES to any **→ Think HEAP ✅**

6. AHA MOMENTS (DON'T FORGET)

- ✅ Heap = Priority, NOT Sorting.
- ✅ Heap guarantees the BEST element only.
- ✅ Ask yourself: "What should I discard?"
- ✅ Heap size K keeps only what I care about.
- ✅ Comparator decides priority (not the data structure).
- ✅ Every Heap problem fits into one of the 4 patterns on **Page 2**.

⭐ **GOLDEN RULE**
The best way to solve Heap problems is to RECOGNIZE them fast, choose the RIGHT HEAP, and MAINTAIN the INVARIANT.

Master Recognition, Not Just Implementation !

HEAP – PATTERNS

Top Interview Patterns with Heaps

Max Heap → largest element on top **Min Heap** → smallest element on top **PQ** → PriorityQueue

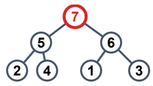
PATTERN 1 REPEATED MAX / MIN RETRIEVAL (e.g., Last Stone Weight)

WHEN TO USE?

- We repeatedly take the largest (or smallest) element.
- Do some operation with it.
- Put the result back.
- Repeat until 1 (or none) left.

HEAP TO USE

Max Heap → largest element on top



IDEA / AHA

Always care about the current largest (or smallest) element.

Heap gives it in O(1).

TEMPLATE / APPROACH (JAVA)

```java
PriorityQueue<Integer> maxHeap =
  new PriorityQueue<>(Collections.reverseOrder());

for (int x : nums) maxHeap.offer(x);

while (maxHeap.size() > 1) {
  int a = maxHeap.poll(); // largest
  int b = maxHeap.poll(); // 2nd largest
  if (a != b) maxHeap.offer(a - b);
}
return maxHeap.isEmpty() ? 0 : maxHeap.peek();
```

EXAMPLES

📌 LC 1046 Last Stone Weight

Complexity
Time: O(n log n)
Space: O(n)

PATTERN 2 TOP-K ELEMENTS (e.g., Kth Largest, Top K Frequent)

WHEN TO USE?

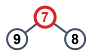
- Need the K largest / smallest / most frequent elements.
- K is much smaller than n.
- We don't want to sort everything.

HEAP TO USE

Top K Largest → Min Heap (size K)

(Keep K=3 largest seen so far)

Top K Smallest → Max Heap (size K)



(Keep K=3 smallest seen so far)

IDEA / AHA

Keep only K best candidates in heap.

If better element comes, replace root (worst among K).

TEMPLATE / APPROACH (JAVA)

Top K Largest (Min Heap of size K)

```java
PriorityQueue<Integer> pq = new PriorityQueue<>();
for (int x : nums) {
  pq.offer(x);
  if (pq.size() > k) pq.poll(); // remove smallest
}
// pq contains k largest elements
```

Top K Smallest (Max Heap of size K)

```java
PriorityQueue<Integer> pq = new PriorityQueue<>(
  Collections.reverseOrder());
for (int x : nums) {
  pq.offer(x);
  if (pq.size() > k) pq.poll(); // remove largest
}
// pq contains k smallest elements
```

EXAMPLES

- LC 215 Kth Largest Element in an Array
- LC 347 Top K Frequent Elements
- LC 973 K Closest Points to Origin
- LC 692 Top K Frequent Words

Complexity
Time: O(n log k)
Space: O(k)

PATTERN 3 MERGE K SORTED LISTS / ARRAYS (e.g., Merge K Sorted Lists)

WHEN TO USE?

- We have K sorted lists / arrays.
- Need to merge into one sorted output.
- Always pick the smallest current candidate.

HEAP TO USE

Min Heap

1→4→7
2→5→8
3→6→9
↓
Min Heap
↓
1 2 3 4 5 6 7 8 9

IDEA / AHA

Put the first element of each list into heap.

Pop smallest, add its next element (if any).

Repeat.

TEMPLATE / APPROACH (JAVA)

```java
class Node {
  int val, list, index; // value, which list, index in list
}
PriorityQueue<Node> pq = new PriorityQueue<>(
  (a, b) -> a.val - b.val);

// 1. Add first element of each list
for (int i = 0, i < lists.length; i++) {
  if (lists[i].size() > 0)
    pq.offer(new Node(lists[i].get(0), i, 0));
}

List<Integer> res = new ArrayList<>();
while (!pq.isEmpty()) {
  Node cur = pq.poll();
  res.add(cur.val);
  int nextIdx = cur.index + 1;
  if (nextIdx < lists.get(cur.list).size()) {
    pq.offer(new Node(lists.get(cur.list).get(nextIdx),
      cur.list, nextIdx));
  }
}
// res is merged sorted list
```

EXAMPLES

- LC 23 Merge K Sorted Lists
- LC 378 Kth Smallest Element in a Sorted Matrix
- LC 1246 Palindrome Removal

Complexity
Time: O(N log k)
(N = total elements)
Space: O(k)

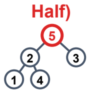
PATTERN 4 TWO HEAPS (MEDIAN FROM DATA STREAM) (e.g., Find Median from Data Stream)

WHEN TO USE?

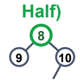
- Need median at any point in a stream.
- Insertions happen one by one.
- Need O(log n) insert and O(1) median.

HEAP TO USE

Max Heap (Smaller Half)



Min Heap (Larger Half)



IDEA / AHA

Maintain two heaps:
1) maxHeap (left half)
→ top = max of left half
2) minHeap (right half)
→ top = min of

TEMPLATE / APPROACH (JAVA)

```java
PriorityQueue<Integer> maxHeap =
  new PriorityQueue<>(Collections.reverseOrder());
PriorityQueue<Integer> minHeap = new PriorityQueue<>();

void addNum(int num) {
  if (maxHeap.isEmpty() || num <= maxHeap.peek())
    maxHeap.offer(num);
  else
    minHeap.offer(num);
  // Balance
```

EXAMPLES

- LC 295 Find Median from Data Ostream

Complexity
Time: O(log n) per operation
Space: O(n)

		right half Balance size difference ≤ 1. Median is either top of maxHeap or avg of both tops.	`if (maxHeap.size() - minHeap.size() > 1)`
			`minHeap.offer(maxHeap.poll());`
			`else if (minHeap.size() > maxHeap.size())`
			`maxHeap.offer(minHeap.poll());`
			`}`
			`double findMedian() {`
			`if (maxHeap.size() > minHeap.size())`
			`return maxHeap.peek();`
			`else`
			`return (maxHeap.peek() + minHeap.peek()) / 2.0;`
			`}`	

🎯 RECOGNITION TRIGGERS (If you see these in a question, think HEAP!)

→ Top K / Kth Largest / Smallest
→ Most Frequent / Top Frequent
→ Merge K Sorted
→ Running Median / Median at any time
→ Continuously pick Max / Min

⭐ **GOLDEN RULE:**
Heaps are all about PRIORITY, not sorting. Choose the right heap and maintain the invariant!